Java Chapter 3: Decisions

Contents

Java Chapter 3: Decisions 1
DRY 1
Do: Think Java, 2nd Ed. (Interactive Edition) 1
Do: Java Tutorials 1
Visualize and Debug Programs 2
If Selection Statement 2
If Else 3
If Else If 3
Compile Java Programs at the Command Line 4
Tutorial 3.1: IsItPositive 4
Tutorial 3.2: IsItPositive2 5
Tutorial 3.3: Grades 6
Logical Operators 8
Tutorial 3.4: Logical Operators 10
Assignment Submission 11

Time required: 90 minutes

DRY

Don't **R**epeat **Y**ourself

Do: Think Java, 2nd Ed. (Interactive Edition)

- Chapter 3 Input and Output
- Chapter 5 Conditionals and Logic

Do: Java Tutorials

- LearnJavaOnline.org Conditionals

- Java Type Casting
- Java Operators
- Java Strings
- Concatenation
- Numbers and Strings
- Special Characters
- Java Math
- Java Booleans
- Java If…Else
- Short Hand If…Else
- Java Switch

Visualize and Debug Programs

The website www.pythontutor.com helps you create visualizations for the code in all the listings and step through those programs one instruction at a time. As you do that, pay attention to the movements of the red and green arrows on the left, the diagram on the right, and the printed material at the bottom. This will help you to better understand the behavior of the programs we are working on.

This is a great way to debug your code. You can see the variables change as you step through the program.

https://pythontutor.com/java.html#mode=edit

If Selection Statement

Same concept, different syntax.

```
public class If {
    public static void main(String[] args) {
        int age = 19;
        if (age >= 16) {
            System.out.println("You can drive!");
        }
    }// end main
```

```
}
```

If Else

```
public class IfElse {
    public static void main(String[] args) {
        int age = 19;
        if (age >= 16) {
            System.out.println("You can drive!");
        } else {
            System.out.println("You cannot drive yet!");
        } // end if-else
    }// end main
}
```

If Else If

```
import java.util.Scanner;
public class IfElseIf {
    public static void main(String[] args) {
        int age;
        Scanner keyboard = new Scanner(System.in);
        System.out.println("Welcome to the Sunset Bar and Grille.");
        System.out.print("Please enter your age: ");
        age = keyboard.nextInt();
        if (age >= 21) {
            System.out.println("Here, have a beer.");
        } else if (age >= 16) {
            System.out.println("Here have a Coke!");
            System.out.println("At least you can drive!");
        } else {
            System.out.println("Here have a Coke!");
        }
        System.out.println("Thanks for coming to the Sunset Bar and
Grille!");
        keyboard.close();
    }
}
```

Compile Java Programs at the Command Line

You may find yourself programming in an environment where you may not have a helpful IDE.

1. Start a command prompt in the folder that contains the Java program.

```
// Compile the code file to HelloWorld.class, the bytecode file
javac HelloWorld.java
// Run the bytecode file
java HelloWorld
```

Tutorial 3.1: IsItPositive

This program uses an **if** statement to determine if we have a positive number.

Create and test the following program called **IsItPositive.java**

```java
// Name: IsItPositive.java
// Written by:
// Written on:
// Purpose: Is the number greater than 0

public class IsItPositive {
    public static void main(String[] args) {

        int number = 10;

        // If number is greater than 0
        if (number > 0) {
            System.out.println(number + " is positive.");
        }

        System.out.println("Statement outside if block");
    }
}
```

Example program run:

```
10 is positive.
Statement outside if block
```

Tutorial 3.2: IsItPositive2

This program uses an **if else** statement to determine if we have a positive or negative number.

Create and test the following program called **IsItPositive2.java**

```java
// Name: IsItPositive2.java
// Written by:
// Written on:
// Purpose: Is the number positive or negative

// Import Scanner library for input
import java.util.Scanner;

public class IsItPositive2 {
    public static void main(String[] args) {
        // Declare Scanner object and initialize with
        // predefined standard input object, System.in
        Scanner keyboard = new Scanner(System.in);

        // Declare integer variable for user input
        int number = 0;

        // Print the heading and prompt
        System.out.println("+-----------------------------------+");
        System.out.println("|--    Is the Number Positive?    --|");
        System.out.println("+-----------------------------------+");
        System.out.print("Enter a number: ");

        // Get integer from the keyboard
        // Assign integer to variable
        number = keyboard.nextInt();

        // If number is greater than 0
        if (number > 0) {
            System.out.println(number + " is positive.");
        } else {
            System.out.println(number + " is negative.");
        }
        keyboard.close();
    }
}
```

Example run:

```
 ----jGRASP exec: java IsItPositive2
+-----------------------------------+
|--    Is the Number Positive?    --|
+-----------------------------------+
Enter a number: 5
5 is positive.

 ----jGRASP: operation complete.

 ----jGRASP exec: java IsItPositive2
+-----------------------------------+
|--    Is the Number Positive?    --|
+-----------------------------------+
Enter a number: -9
-9 is negative.

 ----jGRASP: operation complete.
```

Tutorial 3.3: Grades

Here is a simple use of an **if** statement is to assign letter grades. Suppose that scores 90 and above are A's, scores in the 80s are B's, 70s are C's, 60s are D's, and anything below 60 is an F. Here is one way to do this:

```
# Less effecient decision structure

grade = int(input("Enter your score: "))

if grade >= 90:
    print("A")
elif grade >= 80 and grade < 90:
    print("B")
elif grade >= 70 and grade < 80:
    print("C")
elif grade >= 60 and grade < 70:
    print("D")
else:
    print("F")
```

The code above is straightforward, and it works. A more elegant and simpler way to do it is shown below

Create and save the following program as **Grades.java**

```java
// Name: Grades.java
// Written by:
// Written on:
// Purpose: Is the number positive or negative

// Import Scanner library for input
import java.util.Scanner;

public class Grades {
    public static void main(String[] args) {
        // Declare Scanner object and initialize with
        // predefined standard input object, System.in
        Scanner keyboard = new Scanner(System.in);

        int score;
        String letterGrade;

        // Prompt the user for input
        System.out.println("+-----------------------------------+");
        System.out.println("|--     Please enter a score      --|");
        System.out.println("+-----------------------------------+");
        System.out.print("Enter a score: ");

        // Get integer from the keyboard
        // Assign integer to variable
        score = keyboard.nextInt();

        // Determine what the letter grade is
        if (score >= 90) {
            letterGrade = "A";
        } else if (score >= 80) {
            letterGrade = "B";
        } else if (score >= 70) {
            letterGrade = "C";
        } else if (score >= 60) {
            letterGrade = "D";
        } else {
            letterGrade = "F";
        }

        System.out.println("The score " + score + " is: " + letterGrade);
        keyboard.close();
    }
}
```

Example run:

```
+-----------------------------------+
|--     Please enter a score      --|
+-----------------------------------+
Enter a score: 90
The score 90 is: A

 ----jGRASP: operation complete.

 ----jGRASP exec: java Grades
+-----------------------------------+
|--     Please enter a score      --|
+-----------------------------------+
Enter a score: 59
The score 59 is: F
```

If we use separate **if** statements, each condition is checked regardless of whether it really needs to be. That is, if the score is a 95, the first program will print an A but then continue on and check to see if the score is a B, C, etc., which is a bit of a waste.

Using **else if**, as soon as we find where the score matches, we stop checking conditions and skip all the way to the end of the whole block of statements.

When using **else if**, the second part of the second if statement condition, **grade < 90**, becomes unnecessary because the corresponding **else if** does not have to worry about a score of 90 or above, as such a score would have already been caught by the first if statement.

Logical Operators

A logical operator is a symbol or word that connects two or more expressions so that the value of the produced expression created is solely determined by the value of the original expressions and the operator's meaning.

Following are the logical operators available in Java.

- and
- or
- not

Operator Symbol	**Operator Name**	**Example**	**Description**
&&	Logical And	Operand-A && Operand-B	It returns True, when both the operand-A and operand-B is True, otherwise it returns False.
||	Logical Or	Operand-A || Operand-B	It returns True, when either the operand-A or operand-B is True, otherwise it returns False.
!	Logical Not	! Operand-A	It returns the operand's logical state in reverse.

Tutorial 3.4: Logical Operators

```java
/**
 * Name: LogicalOperators.java
 * Written by:
 * Written on:
 * Purpose: Demonstrate combinations of and, or, not
 */

class LogicalOperators {
    public static void main(String args[]) {

        int A = 10;
        int B = 20;
        int C = 30;

        // And Operator
        // Both True
        System.out.print("(A < B && B < C): ");
        System.out.println((A < B && B < C));
        // First operand: True Second Operand: False
        System.out.print("(A < B && B > C): ");
        System.out.println((A < B && B > C));
        // Both False
        System.out.print("(A > B && B > C): ");
        System.out.println((A > B && B > C));

        System.out.println();

        // OR operator
        // Both True
        System.out.print("(A < B || B < C): ");
        System.out.println((A < B || B < C));
        // First operand: True Second Operand: False
        System.out.print("(A < B || B > C): ");
        System.out.println((A < B || B > C));
        // Both False
        System.out.print("(A > B || B > C): ");
        System.out.println((A > B || B > C));

        // Not operator
        boolean D = true;
        // reverse of true logical state
        System.out.print("(!D): ");
        System.out.println((!D));

    }
}
```

Example run:

```
(A < B && B < C): true
(A < B && B > C): false
(A > B && B > C): false

(A < B || B < C): true
(A < B || B > C): true
(A > B || B > C): false
(!D): false
```

Assignment Submission

1. Attach the pseudocode.
2. Attach the program files.
3. **Command Line Compile:** Attach screenshots showing the successful command line compile and run of one program.
4. Submit in Blackboard.